

ELECTIVE COURSE-1.1

UCS546: CONVERSATIONAL AI: ACCELERATED DATA SCIENCE [BASICS]				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: This course will provide students with fundamental knowledge of Data Science Pipeline from Data Collection to Analytics using RAPIDS and Spark Framework.				
Syllabus Introduction: Fundamentals of Data Science, GPU Acceleration, Pipeline: Data Collection, Preprocessing, Analytics and Visualization, Working with Big Data. Fundamentals of Data Analytics & Data Science: What is Data Science and Why is Data Science Importance, Analytical Building Blocks, Examples Data Science Projects 1: Apollo Graph Exploration, Project 2: NetProbe Auction Fraud Detection, Data Science Buzzwords, Hype Cycle, General vs Narrow AI. Data Collection: Collecting Data, Scraping Data, Popular Scraping libraries, Data Annotation and Data Quality, SQLits as Simple, SQL Refresher, Missing Indexes. Data Pre-processing (ETL): Introduction to Data-preprocessing, Data Cleaning & Statistical Preprocessing, Data Cleaners: OpenRefine and Wrangler, Feature Selection: Introduction to Filter Methods, Introduction to Model- based methods, Feature Reduction: PCA. Data Analytics and Visualization: Bias and Fairness Measures, Tools for Discovering & Interpreting Bias in Models, Knowledge Graph, Data deduplication, What is Info Vis, Exploratory Data Analytics, Fixing Common Visualization Issues. Distributed Computing: Hadoop, Spark, Dask and UCX: Big data is common, How to store Big data, Why Hadoop, MapReduce Overview, Pig & Hive, Spark Overview, Spark SQL and other libraries, RAPIDS & Spark Integration, HBase Overview & Usage, Dask and UCX with RAPIDS and BlazingSQL				
Laboratory Work (if applicable) • Introduction to Dockers & Containers, Introduction to NVIDIA GPU Cloud (NGC). • Practical on Traditional Data Science packages (Numpy, Pandas, Scipy, Scikit-Learn). • Accelerated Data Science framework RAPIDS: Introduction to RAPIDS and cuDF. • Data Collection via API. • Data Annotation in Active Learning. • GPU-accelerated SQL with BlazingSQL. {DLI Online Course Section: Fundamentals of Accelerated Data Science with RAPIDS, Section 1: GPU-accelerated Data Manipulation} • Data Wrangling with OpenRefine. • Outlier Detection with IQR. • Feature Reduction with PCA. • Visual Exploratory Data Analytics using cuXFilter. • Web-based Data Visualization. • Server and Client-side Visualizations (Datashader, Plotly, Plotly Dash). • Working with Hadoop and Apache Spark 3.0 for Big Data Analytics.				
Course Learning Outcomes (CLOs) The students will be able to:				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

1. understand GPU computing for building data analytics pipelines.
2. comprehend and apply different association mining, cleaning and analytics techniques.
3. techniques.
4. in depth understanding of Data Analytics, Pre-processing and Visualization Toolkits with optimized acceleration using RAPIDS Framework
5. handle various real time applications through Hadoop and Apache Spark 3.0

Text Books

1. An Introduction to Probability and Statistics by V.K. Rohatgi & A.K. Md.E.Saleh.
2. Introduction to Probability and Statistics by J.S. Milton & J.C.Arnold.
3. Big Data: A Revolution That Will Transform How We Live, Work, and Think, V. Mayer-Schönberger and K. Cukier.

Reference Books

1. A First Course in Probability by S.M. Ross.
2. Probability and Statistics in Engineering by W.W. Hines, D.C. Montgomery, D.M. Gpldsman & C.M.Borrer.
3. Lectures in Probability by M.Kac (for example on independent events).
4. C.K. Wong (1972) A note on mutually independent events. Annals of Statistics, V. 26, 27 (for example on independent events).
5. Measure Theory by P. Halmos (for algebra of sets).

Evaluation Scheme:

Sl. No	Evaluation Elements	Weightage (%)
1	MST	35
2	EST	35
3	Sessional (May include Assignments/Projects/Tutorials/Quiz/Lab evaluations)	30

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